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SARDAR PATEL UNIVERSITY

Bachelor of Science (Semester 5) Examination OCTOBER - 2025

Subject Code: US05MABIC01

Subject Title: Molecular biology

Date: 04/10/2025, Saturday

Time: 10:00 am to 12:30 pm

Total: 50 Marks

NOTE:

1. Figure to the right indicate full marks of the questions.
2. Draw figure when necessary.

Q-1 MULTIPLE CHOICE QUESTIONS

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1. Which of the following is absent in prokaryotic genome?
 - a) Introns
 - b) Operons
 - c) Exons
 - d) Promoter genes
2. Viral genome can be _____
 - a) Only ds DNA
 - b) Only circular DNA
 - c) Only ss DNA
 - d) DNA or RNA, ss or ds
3. Which of the following enzymes separates the two strands of DNA during replication?
 - a) Helicase
 - b) Topoisomerase
 - c) Gyrase
 - d) DNA polymerase
4. Termination of replication is triggered by _____
 - a) DNA polymerase
 - b) SSB
 - c) Helicase
 - d) Tus protein
5. DNA replication is _____
 - a) conservative and discontinuous
 - b) semi-conservative and semi-discontinuous
 - c) conservative
 - d) semi-conservative and discontinuous
6. The direction of RNA synthesis is always:
 - a) 3' to 5'
 - b) Parallel to the DNA template
 - c) 5' to 3'
 - d) Dependent on the DNA sequence
7. If the template strand of DNA has the sequence 3'-TAC-5', what will be the sequence of the newly synthesized RNA molecule in the corresponding region?
 - a) 5'-AUG-3'
 - b) 3'-UAC-5'
 - c) 5'-UAC-3'
 - d) 3'-AUG-5'
8. In prokaryotes, translation of mRNA can begin _____
 - a) Only after transcription is fully completed and the mRNA is released
 - b) Simultaneously with transcription, even before the mRNA is fully synthesized
 - c) In a separate cellular compartment
 - d) Only after the mRNA undergoes splicing

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(P.T.O.)

9. In which of the following directions does the polypeptide synthesis proceeds?
- a) From both N-terminus and C-terminus
 - b) Either from N-terminus or C-terminus
 - c) C-terminus to N-terminus
 - d) N-terminus to C-terminus
10. The second and further aminoacyl-tRNAs are brought to the ribosome bound to which of the following protein complex?
- a) EF-Ts-ATP
 - b) EF-Tu-GTP
 - c) EF-Ts-GTP
 - d) EF-Tu-ATP

Q-2 Answer the following:

- (A) Define: gene and write a note on concept of gene.
- (B) Explain: salient features of Eukaryotic genome.

OR

- (A) Give an account on salient features of viral genome.
- (B) Write a note on: Trisomy.

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Q-3 Answer the following:

- (A) Explain: DNA replication is semi-conservative.
- (B) Write a note on 5'-3' polymerization.

OR

- (A) Explain: Initiation of replication in prokaryotes.
- (B) Describe in detail: Okazaki Fragment

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Q-4 Answer the following:

- (A) Define: transcription and also explain initiation of transcription in prokaryotes.
- (B) Write a note on generation of 5' cap.

OR

- (A) Describe in detail: termination of transcription in prokaryotes.
- (B) Explain: Poly A tail formation.

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Q-5 Answer the following:

- (A) Describe in detail: Post translational modifications.

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OR

- (A) Give an account on inhibition of protein synthesis by antibiotics.
- (B) Write a note on activation of amino acids